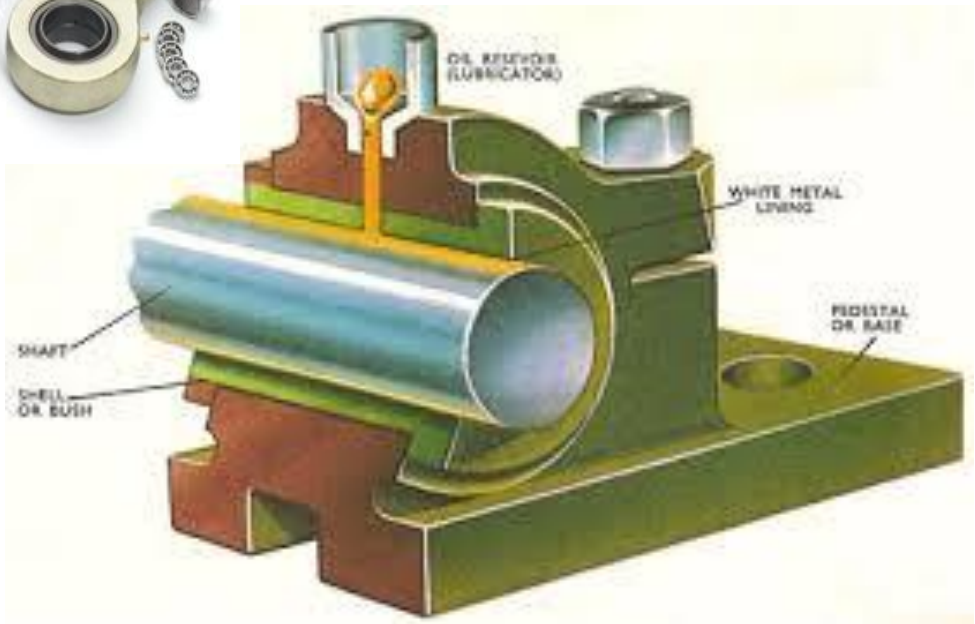
BEARINGS




OIL RESERVOIR (LUBRICATOR)
WHITE METAL LINING
SHAFT
SHELL OR BUSH
PEDESTAL OR BASE

- Applications:



Lead screw



Rotary table bearing



Guideways



Thrust axis vibration of Rocket Payload



Dynamic balancing of rocket payload





Large binocular telescope(LBT)

BEARING

DEFINITION:

- A bearing is a system of machine elements whose function is to support a second member, preventing its motion in the direction of an applied load, at the same time allowing motion in another predetermined direction.

the part of a shaft or axle in contact with or enclosed by a bearing

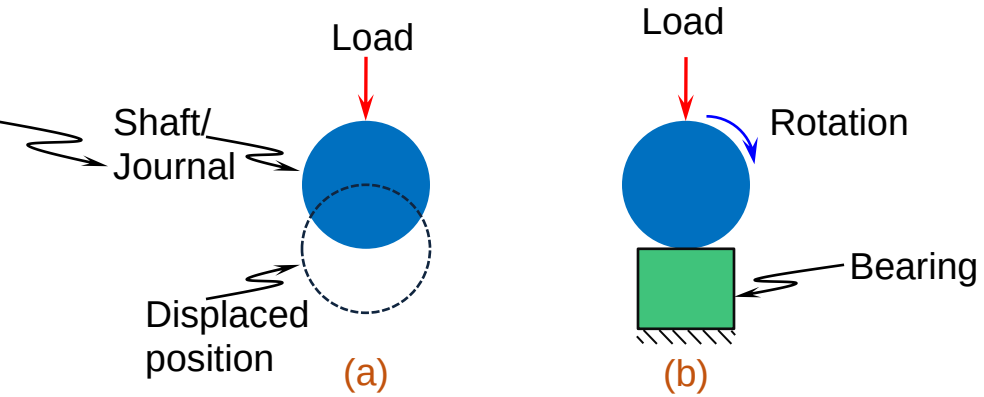


Fig. 9 End views of machine elements (a) without support and (b) with support

Bearings are classified according to the direction of applied load.

- If the bearing supports a radial load, it is called radial or shaft or sliding or journal bearing.
- If the bearing supports a axial or thrust load, it is called thrust bearing.
- Some bearings can support both radial and axial loads and they are known as conical bearings.

TYPES OF BEARINGS:

- There are two main types of bearings commonly used in practice and are;

  (a) Rolling contact bearing

  (b) Sliding or Fluid-film bearing

- Rolling contact bearings are bearings in which balls, rollers or needles are used to support the applied load.

- Since rolling friction is lower than the sliding friction, rolling contact bearings have much wider use in industries.

- If the bearings are operating on the principle of hydrodynamic lubrication condition, such bearings are called fluid-film bearings.

- Fluid-film bearings are further categorized as;
  - (i) Self-acting or hydrodynamic bearings
  - (ii) Externally pressurized or hydrostatic bearings
  - (iii) Squeeze-film or elasto-hydrodynamic bearing

- A fluid-film bearing operating on the hydrodynamic condition and the load is supported due to wedging effect of the fluid caused by the relative tangential motion between two surfaces, such bearings are called self-acting bearings.

- In an externally pressurized (some times called hydrostatic) bearing, the load is supported due to pressure in the fluid which is supplied from an external source.

- In the case of squeeze-film bearing, which supports a load because of oscillating relative normal motion.

- In the last two categories i.e., externally-pressurized and squeeze-film bearings, the pressure within the clearance space can be generated without the wedging effect.

- Fig. 10 depicts the types of the bearings.

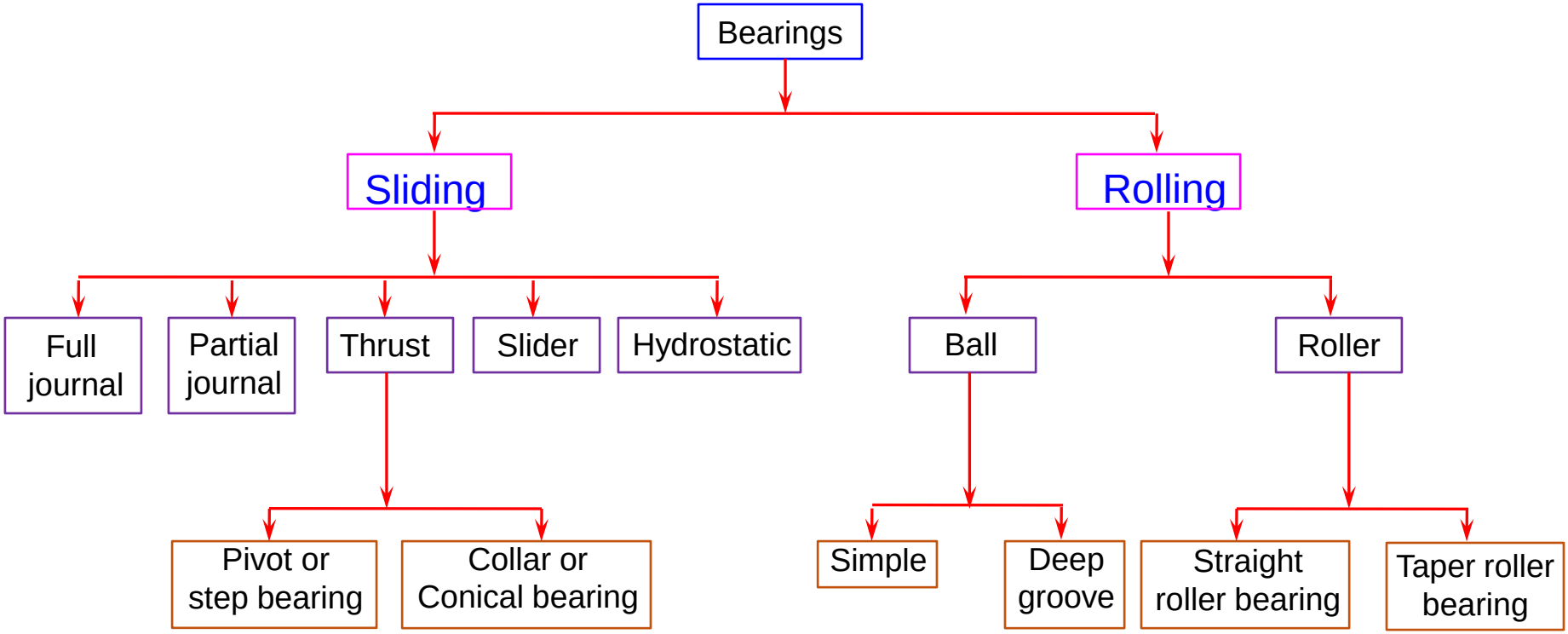


Fig. 1 Types of bearings

Journal Bearings

- Journal or shaft bearings which supports radial load may be divided into full-journal bearing, where the contact angle of the bushing with the journal is 360° and the partial-journal bearing in which the contact angle is either 180° or less.
- The shaft member of this bearing unit is called journal and the cylindrical body around the journal is called bearing.
- In most installations, the journal is a rotating element and the bearing element is stationary. Some times, both elements are allowed to rotate.

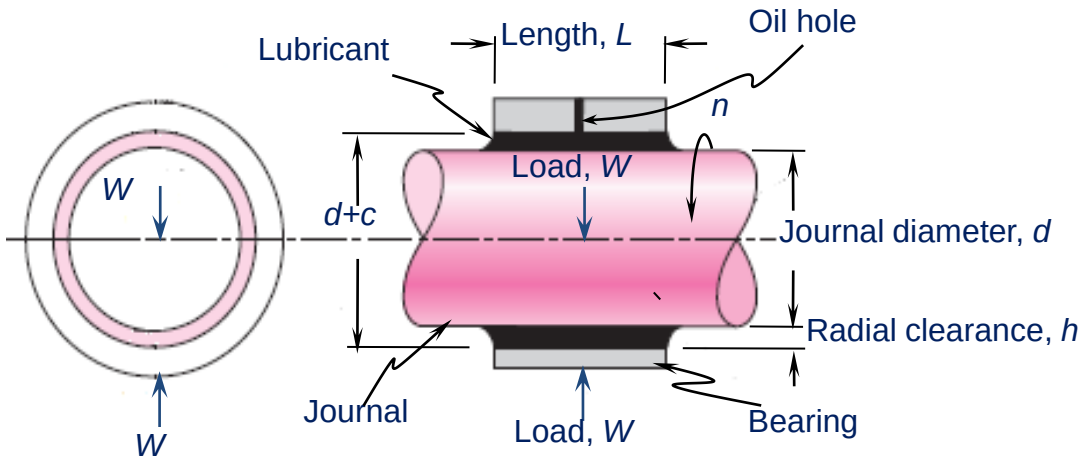


Fig. 2 Schematic of full-journal bearing

Fig. 3 represents a plane thrust bearing, in which journal and bearing have plane rubbing surfaces provided and thus the bearing operates under condition of boundary lubrication.

This type of bearings able to support only a relatively light load at moderate speeds.

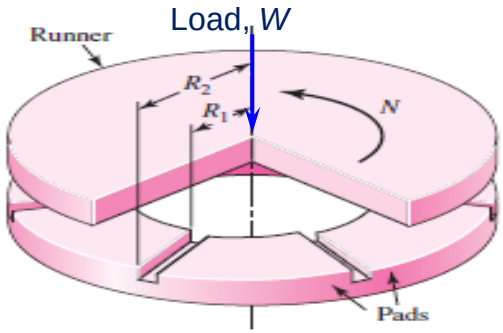
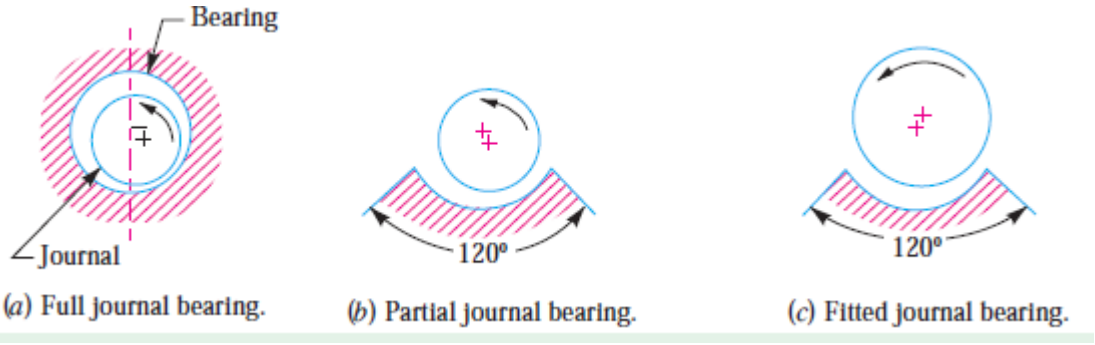


Fig. 3 Schematic of Thrust bearing



(a) Full journal bearing. (b) Partial journal bearing. (c) Fitted journal bearing.

Derivation of Petroff's Equation

- Petroff's equation is used to determine the coefficient of friction in journal bearing.
- It is based on the following assumptions:
  - Shaft is concentric with the bearing
  - Bearing is subjected to light load
  - Clearance is completely filled with lubricant
  - Journal revolves at high speeds
  - No end leakage.
- Consider a journal bearing as shown in Fig. 4.

Let c - diametral clearance

d - diameter of journal

n'- speed of journal in rps=N/60

L - length of bearing

$\Psi = \frac{c}{d}$ - diametral clearance ratio

v- velocity $= \frac{\pi dn}{60} = \pi dn'$ in m/s

η- viscosity of lubricant in Pas.

W- normal load

μ - Coefficient of friction

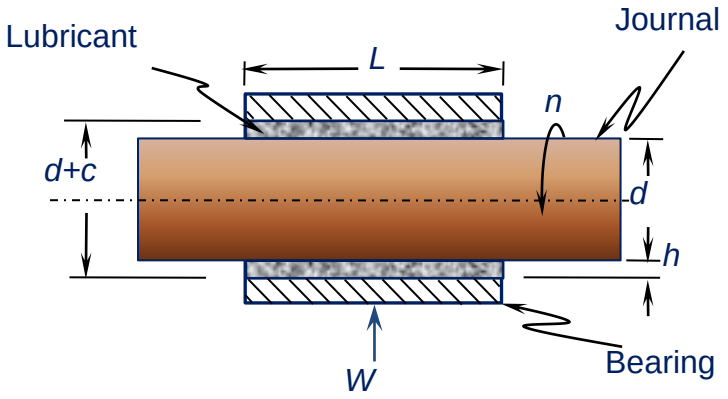


Fig. 4 Schematic of journal bearing

Conti….

- Shear stress, $\tau = \eta \frac{v}{h}$ $\because$ Abs. viscosity $\eta = \frac{\tau}{(v/h)}$ N-s/m^2 or Pa-s

- Substituting values of ***v*** and ***h*** we get,

$$\tau = \eta \frac{\pi d n'}{c/2} = \frac{2\pi d n' \eta}{c}$$

- Surface area, $A = \pi d L$

- Frictional or shear force, $F = \tau A = \frac{2\pi d n' \eta}{c} \times \pi d L = \frac{2\pi^2 d^2 n' \eta L}{c}$

- Frictional torque, $M_t = F \frac{d}{2}$

$$= \frac{2\pi^2 d^2 n' \eta L}{c} \times \frac{d}{2}$$

$$= \frac{\pi^2 d^2 n' \eta L}{(c/d)}$$

$$= \frac{\pi^2 d^2 n' \eta L}{\Psi} \quad \cdots(1)$$

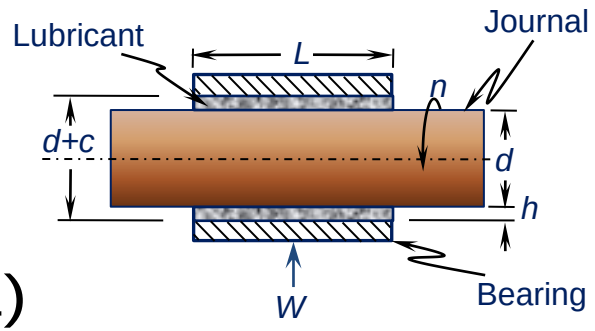


Fig. 4 Schematic of journal bearing

- But frictional torque, $M_t = (\mu W)\frac{d}{2}$
- Normal load, $W = pA_P$
  where p- bearing pressure and A_p is projected area=Ld.

$$\therefore \quad W = p(Ld)$$

$$\therefore \quad M_t = (\mu pLd)\frac{d}{2} = \mu\frac{pLd^2}{2} \qquad \cdots(2)$$

- Equating equations (1) and (2) we get, the coefficient friction as

$$\frac{\pi^2 d^2 n' \eta L}{\Psi} = \mu\frac{pLd^2}{2}$$

$$\therefore \quad \mu = 2\pi^2\left(\frac{\eta n'}{p}\right)\frac{1}{\Psi} \qquad \cdots(3)$$

- Equation (3) is called Petroff's equation.

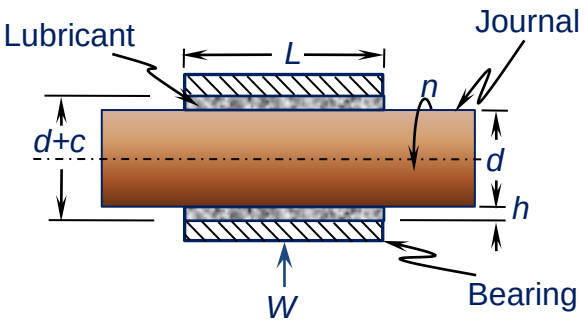
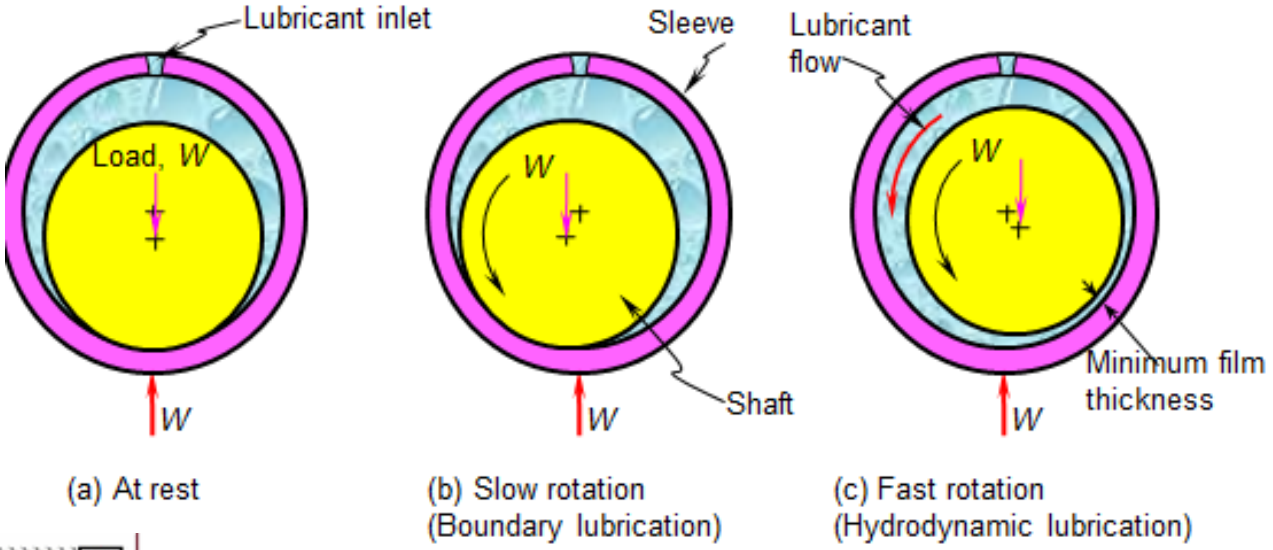


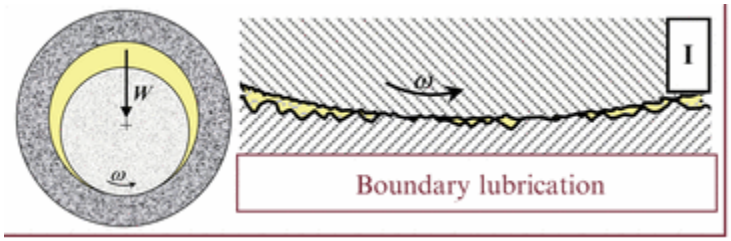
Fig. 4 Schematic of journal bearing

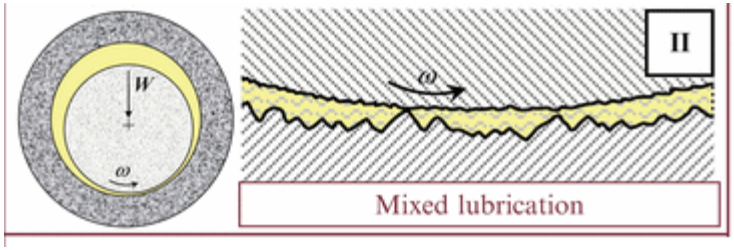
McKee's Equation

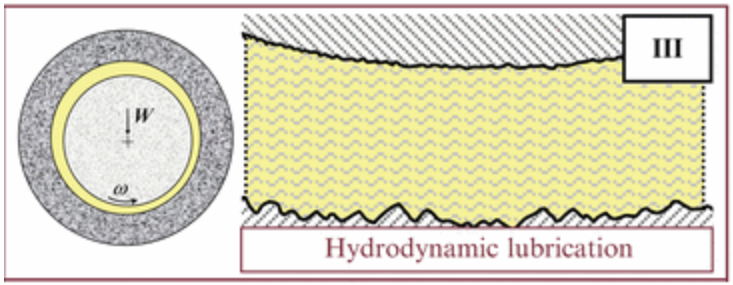
- In hydrodynamic bearings, initially the journal is at rest, thus there is no relative motion and no hydrodynamic film
- Therefore, there is metal to metal contact between surfaces of the journal and the bearing.
- As the journal starts to rotate, it takes some time for the hydrodynamic film to build sufficient pressure in the clearance space.
- During this period, there is partial metal to metal contact and partial lubricant film and this is known as thin film lubrication.
- As the speed increased, more and more, lubricant forced to into wedge shaped clearance space and sufficient pressure is built up, separating the surfaces and this is thick film lubrication .
- Therefore, there is a transition from thin film lubrication to thick film lubrication as the speed increases.

Hydrodynamic lubrication









- The transition form was first observed by McKee brothers in the experimental study through means of a curve called Stribeck curve as shown in Fig. 5.
- Referring to Fig. 5:
  - In region *BC*, there is partial metal to metal contact and partial patches of lubricant. This is the condition of thin film or boundary lubrication
  - In region *CD*, there is relative thick film of lubricant and hydrodynamic lubrication takes place.
  - *AC* is the dividing line between these two modes of lubrication.
  - The coefficient of friction is minimum at '*C*', the value of bearing characteristic number, ($\eta n'/p$). Corresponding to this minimum coefficient is called bearing modulus and it is denoted by κ_a in the Figure.

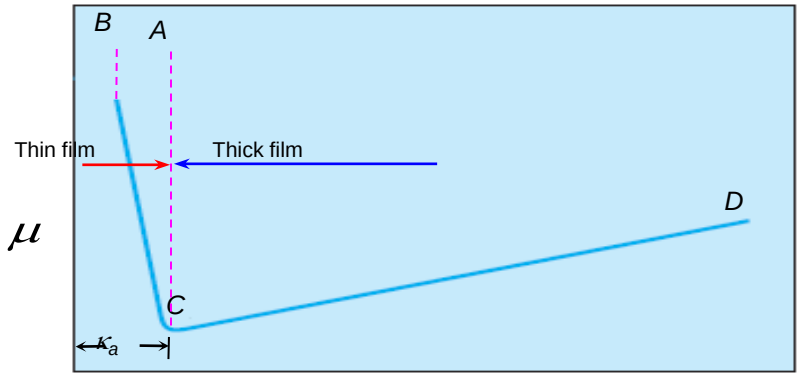


Fig. 5. Stribeck curve

- The bearing should not be operated near the critical value κ_a at point *C*.
- Slight drop in speed *n′* or slight increase in load, will reduce the value of $\eta n'/p$.
- The coefficient of friction for a well lubricated journal bearing is given by

$$\mu = \kappa_a \left(\frac{\eta n'}{p}\right)\left(\frac{r}{c}\right)10^{-10} + \Delta\mu \quad \cdots(4) \qquad \mu \text{ or } f = \frac{0.326}{10^6}\left(\frac{Zn}{p}\right)\frac{D}{C_d} + K_f$$

where, $\kappa_a = 541.33\,\beta$ *where circumferential length of bearing in deg*

$= 0.195\times10^6$ for full journal bearing i.e., $\beta = 360^0$

$\Delta\mu$- factor to correct for end leakage=0.002. for bearing having

$$0.75 < \frac{L}{D} < 2.8$$

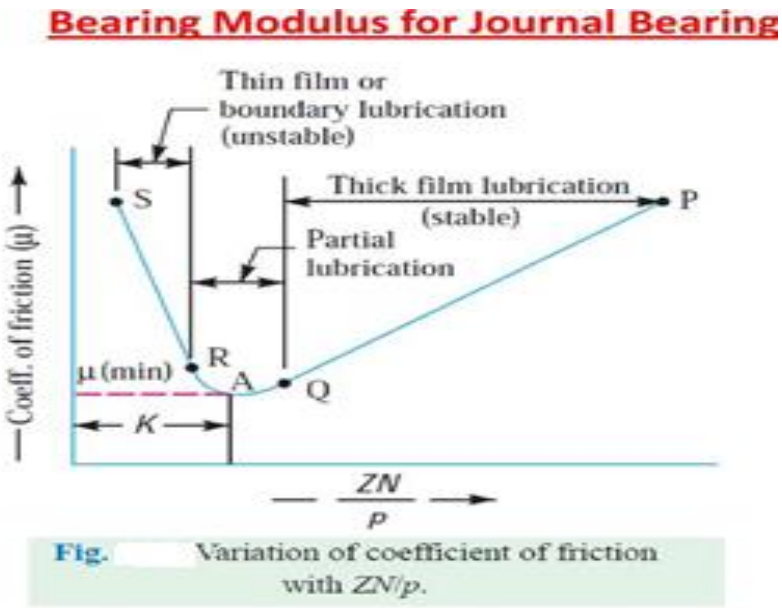


Fig. Variation of coefficient of friction with *ZN/p*.

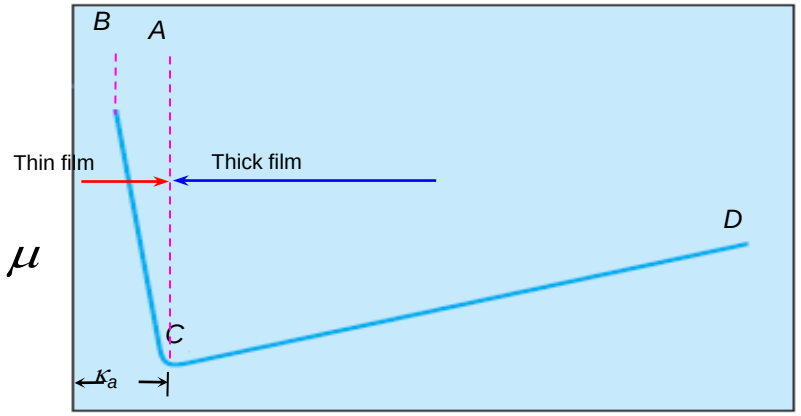
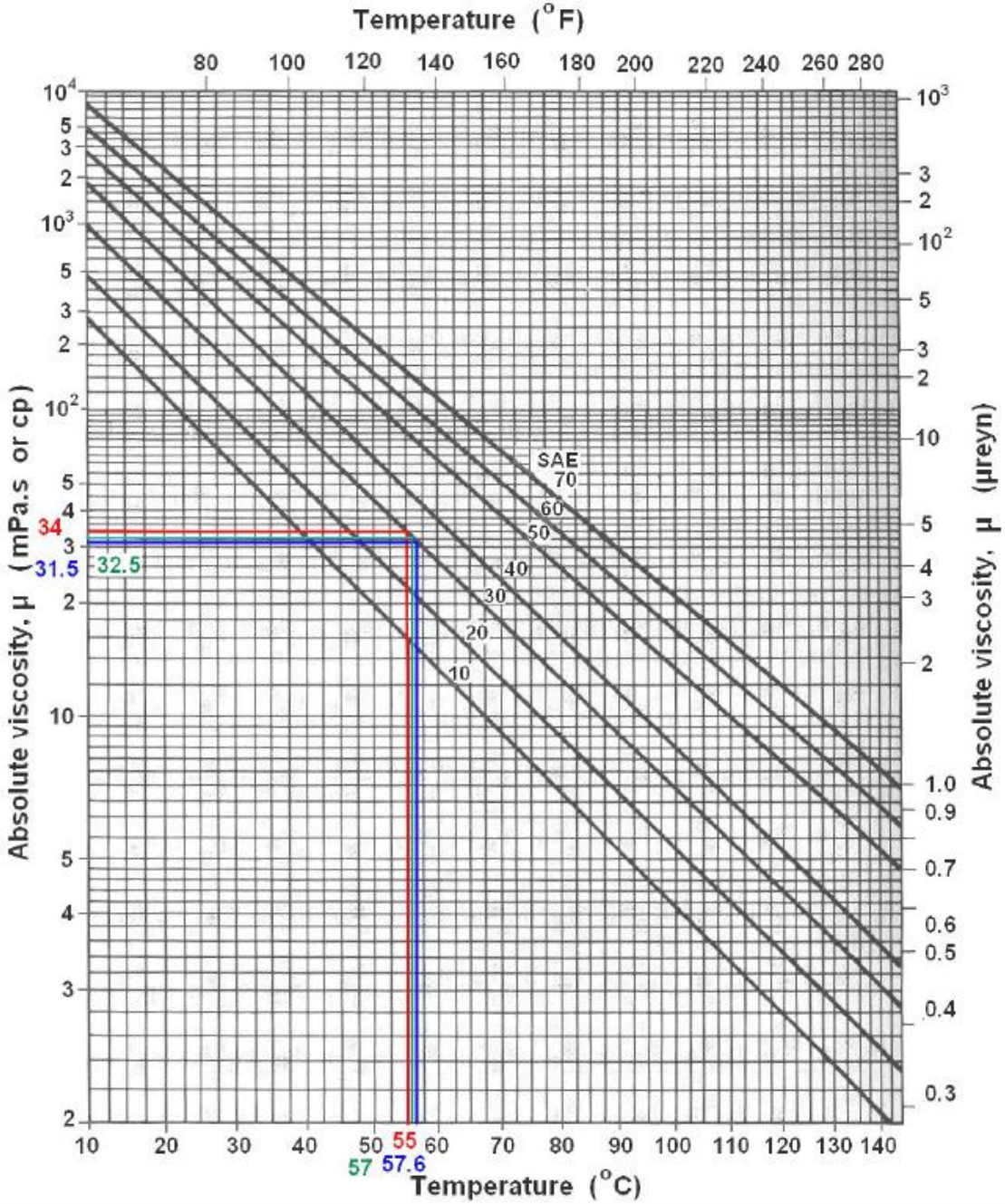
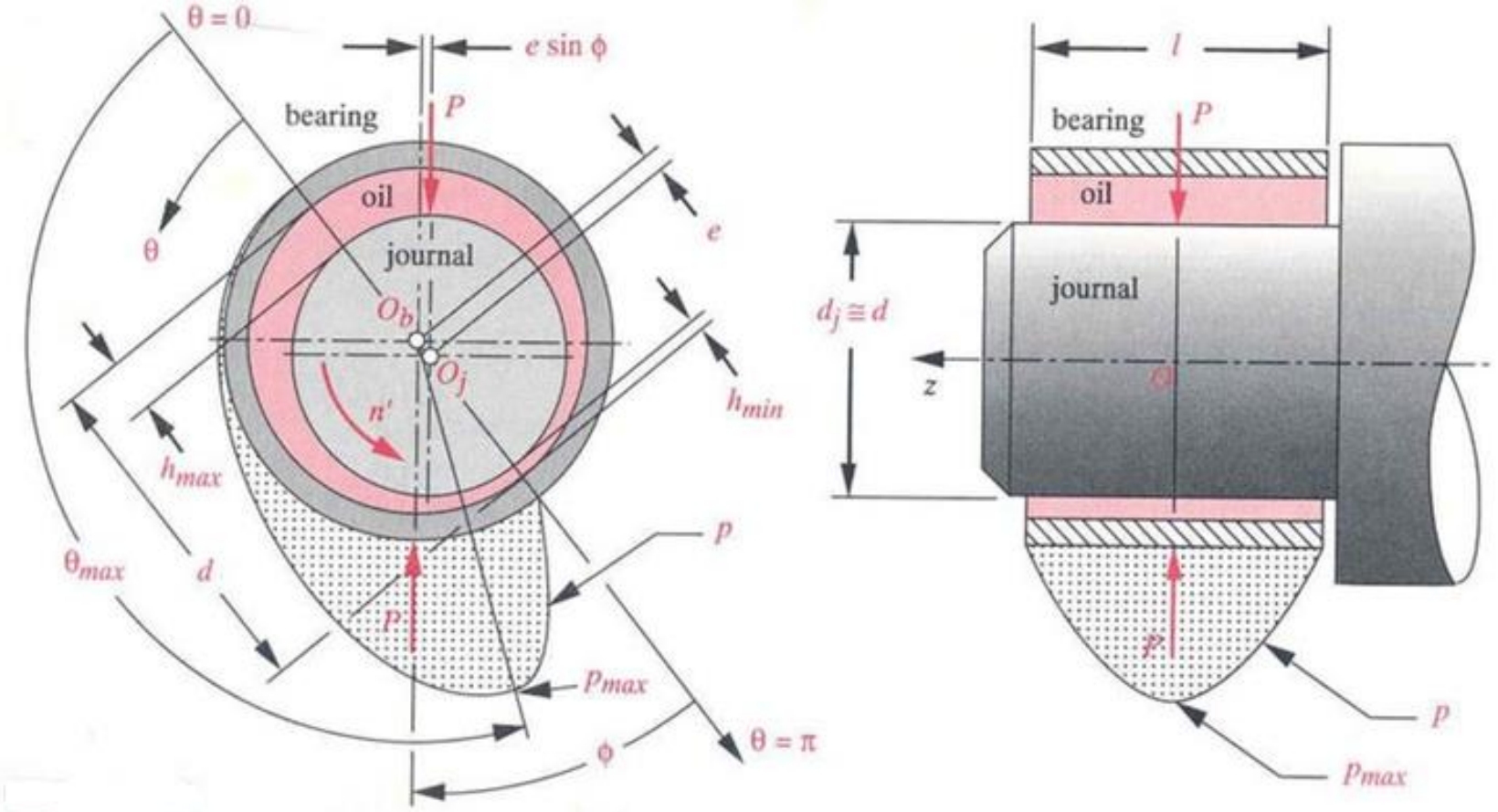


Fig. 5. Stribeck curve



Viscosity – temperature curves of SAE graded oils

Terminology



Terminology

D = diameter of bearing

d = diameter of journal

L = length of bearing or journal

L/D = 1 square bearing

L/D > 1 = for long bearing

L/D < 1 = for short bearing

Diametral clearance C_d =D-d

Radial Clearance C_r =(D-d)/2

Diametral clearance ratio Ψ = C_d /d

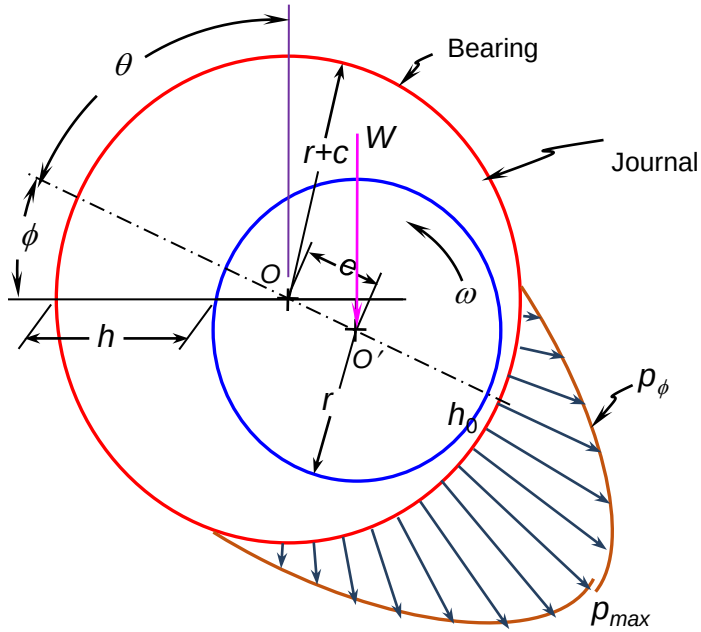


Fig.6 Pressure distribution in Journal bearing

Eccentricity e: Distance between center of journal and center of bearing

Attitude or eccentricity ratio ε = e/ C_r = 2e/ C_d b'cause (C_r =C_d/2)

Also, ϵ =1-2h_{min}/C_d also, h_{min} = h_0

ϵ = 0, the journal runs concentric with bearing

E = 1, the journal touches the bearing

Terminology

Average Pressure P = Load / Projected Area = W / L D

Bearing modulus C = Zn/P where Z is abs viscosity, n speed in rpm, P ave pressure

Average value of bearing modulus should be Zn/P ≥ 3C

For larger fluctuations and heavy impact loads, Zn/P = 15 C

Sommerfeld number or Bearing characteristic number

$$S = \left(\frac{\eta n'}{p}\right)\left(\frac{1}{\Psi^2}\right) \quad \text{or} \quad \left(\frac{Zn'}{p}\right)\left(\frac{D}{C_d}\right)^2$$

Where n' speed in rps

$$\textit{Operating pressure } P = \frac{zn}{4.75\times10^6}\left(\frac{D}{C_d}\right)^2\left(\frac{L}{D+L}\right) \quad \text{MN/m}^2$$

$$\textit{Coefficient of friction } \mu \textit{ or } f = \frac{0.326}{10^6}\left(\frac{Zn}{p}\right)\frac{D}{C_d} + K_f$$

Where Kf= factor to correct for end leakage. Its value varies with L/D ratio

$$0.75 < \frac{L}{D} < 2.8$$

- For analysis of an idealized journal bearing, following assumptions are made:

a) Lubricant obeys Newton's viscous effect.

b) Forces due to the inertia of the lubricant are neglected.

c) Lubricant is assumed to be incompressible.

d) Viscosity is assumed to be constant throughout the film.

e) Pressure does not vary in the axial direction.

f) There is no flow in an axial direction.

g) Flow of the lubricant in the film is laminar.

h) There is no slip between the lubricant and bearing surfaces.

HEAT BALANCE IN AN IDEALIZED FULL JOURNAL BEARINGS

Heat Generation:

- Heat generated in an idealized full journal bearing due to the friction is given by

$$H_g = \mu W v = \mu p L d v \quad J/s \text{ or Watt} \quad \cdots(43)$$

- where $v = \frac{\pi d N}{60}$, $p = p_{out}$, $\mu = \kappa_a \left(\frac{\eta n'}{p} \right) \frac{1}{\psi} \times 10^{-10} + \Delta\mu$

and $\kappa_a = 1.95 \times 10^{11}$ for $\beta = 360°$ and $\Delta\mu = 0.002$

- The heat generated can also be found by knowing the temperature rise of lubricant which is used to carry away heat generated in the bearing.

$$H_g = \gamma\, c_{sp} Q\, \Delta T \quad \cdots(44)$$

where γ- Weight per unit volume of lubricant in N/m^3

C_{sp}–Specific heat of lubricant in J/NK

Q – Flow rate of lubricant in m^3/s

ΔT- Difference between outlet and inlet temperature of the lubricant in K

Heat Dissipation:

- Heat dissipated in an idealized full journal bearing is given by

$$H_d = CA(T_b - T_a) \quad \text{Watt} \quad \cdots(45)$$

where *C*- Combined coefficient of radiation and convection = 11.36 W/m^{2}K

A–Effective surface area of bearing housing =25 in m^2

T_b –Temperature of bearing in °C

T_a- Ambient or air temperature in °C

$$T_b - T_a = \frac{T_o - T_a}{2}$$

T_o – Operating temperature in °C

- An artificial cooling unit is necessary if following condition satisfies:
-

$$H_g > H_d \quad \cdots(46)$$

- Otherwise not necessary.

Pressure Fed Bearings

Amount of heat removed by circulating oil

$H_d = \rho c Q (T_o - T_e)$ Watts

where, ρ = density of oil = 900kg/m

c = specific heat of oil =1.84 to 2.05 kj/kg^{o}C

Q = quantity of oil m^3/sec

$t_o - t_e = (outlet\ oil\ temperature - entrance\ oil\ temperature)\ ^0C$

Design Considerations in Hydrodynamic Bearings

- In the design of hydrodynamic bearings, we may distinguish between two groups of variables.
- In the first group are those whose values either are given or are under the control of the designer. These are:
    a) The viscosity, η
    b) The load per unit of projected bearing area, P
    c) The speed, N
    d) The bearing dimensions r, c, β and L.
- Of these four variables, the designer usually has no control over the speed, because it is specified by the overall design of the machine.
- Sometimes the viscosity is specified in advance, as, for example, when the oil is stored in a sump and is used for lubricating and cooling a variety of bearings.
- The remaining variables, and sometimes the viscosity, may be controlled by the designer and are therefore the decisions the designer makes.
- When these four decisions are made, the design is complete.

- In the second group are the dependent variables.
- The designer cannot control these except indirectly by changing one or more of the first group. These are:

    a) The coefficient of friction, μ

    b) The temperature rise, T

    c) The volume flow rate of oil, Q

    d) The minimum film thickness, h_0
- This group of variables tells us how well the bearing is performing, and hence we may regard them as performance factors.
- Certain limitations on their values must be imposed by the designer to ensure satisfactory performance.
- These limitations are specified by the characteristics of the bearing materials and of the lubricant.
- The fundamental problem in bearing design, therefore, is to define satisfactory limits for the second group of variables and then to decide upon values for the first group such that these limitations are not exceeded.

Ex 1: A bearing of 0.5 m in diameter and 0.75 m in length supports a shaft running at 900 rev/min. The room temperature at 32°C. The viscosity of oil used is 0.0128 kg/ms at the operating temperature of 132°C. The diametral clearance is 0.05 mm and the bearing is to operate in still air without any artificial cooling. Determine (i) the permissible load on bearing (ii) power loss